INCH-POUND

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MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M53 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M53 Cartridge Actuated Initiator Remanufactured Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109 Quality Assurance Terms And Definitions
MIL-STD-413 Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

10520621	Initiator, Cartridge Actuated, M53, Metal Parts Assembly
8830994	Box, Fiberboard (Inner), Packing, Ammunition, For Initiator, Cartridge Actuated, M53
8830995	Box, Packing, Ammunition, For Initiator, Cartridge Actuated, M53
9217624	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Initiator, Cartridge Actuated, M53

U.S. Air Force

TPO 00-731-9272	Initiator, Cartridge Actuated, M53

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured initiator metal parts assembly shall contain a new shear pin and new O-rings.

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3.2 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.3.1).

3.3 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration. Remanufactured exterior components (see 6.3.2) shall have the existing protective finish stripped and the specified protective finish reapplied.

3.4 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using the procedures of 4.5.3.

3.5 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 20 ± 1 pounds is applied at the steady rate of 0.125 inch per minute.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.6.1 Integrated plant products. When remanufacturing and loading of the assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.6.2 Threaded component parts. Class 2A or 2B thread gage shall be used on remanufactured component parts inspection versus class 3A or 3B thread gage required on new hardware for initiators.

3.7 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or

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purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77°F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination. On reprocessed parts, one hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for

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rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.3.2.1 Examination of reclaimed parts. Examination of reusable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Filter, Drawing 10520414, covering a detail of Drawing 10520621.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Nylon patch not properly located or fused	Manual/Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr	Visual
105	Foreign matter	Visual
Minor		
None defined		

4.3.2.3 Chamber, Initiator, Drawing 10520407, covering a detail of Drawing 10520621.

Critical		
None defined		
Major		
101	Pitch diameter of large external thread, min	Gage
102	Pitch diameter of small external thread, min	Gage
103	Pitch diameter of internal thread, max	Gage
104	Thread damaged	Visual
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Surface finish on angle of gas port improper	Visual
108	Burr	Visual
109	Foreign matter	Visual

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Minor
201 Surface finish improper, except as otherwise classified — Visual

4.3.2.4 Cap, Drawing 10520473, covering a detail of Drawing 10520621.

Critical
None defined

Major
101 Pitch diameter of all threads, max — Gage
102 Thread damaged — Visual
103 Metal defective — Visual
104 Protective finish inadequate or defective — Visual
105 Burr — Visual
106 Foreign matter — Visual

Minor
None defined

4.3.2.5 Pin, Shear, Drawing 10520622, covering a detail of Drawing 10520621.

Critical
None defined

Major
101 Diameter, max — Gage
102 Total length, min — Measure
103 Metal defective — Visual
104 Protective finish inadequate or defective — Visual
105 Surface finish improper — Visual
106 Burr — Visual
107 Foreign matter — Visual

Minor
201 Total length, max — Gage
202 Profile of end — Gage

4.3.2.6 Pin, Firing, Drawing 10520474, covering a detail of Drawing 10520621.

Critical
None defined

Major
101 Metal defective — Visual
102 Protective finish inadequate or defective — Visual

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103	Surface finish of body and striker tip improper		Visual
104	Burr	Visual	
105	Foreign matter	Visual	

Minor
None defined

4.3.2.7 Assembly, Drawing 10520621, less all Drawings except 10520557, 10520474, and MS29513-9.

Critical

1	Firing pin protrusion		Gage
2	O-ring missing		Visual

Major

101	Burr	Visual	
102	Foreign matter	Visual	

Minor

201	O-ring not lubricated	Visual	

4.3.2.8 Assembly, Drawing 10520621, less Drawing 10520557.

Critical

1	Any part missing, except as otherwise classified		Visual

Major
None defined

Minor

201	Protective cap or plug missing		Visual
202	Protective finish inadequate or defective		Visual

4.3.2.9 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8830994.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual	
202	Box damaged	Visual	
203	Marking missing, incorrect, or illegible		Visual

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204	Cut, damaged, or improperly sealed bag	Visual

4.3.2.10 Box, Packing, Ammunition, Drawing 8830995.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.3.2.11 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217624.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loose, broken, or improperly engaged	Visual
204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect, or illegible	Visual

4.4 Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.1 Hydrostatic pressure. Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.2 shall be cause for rejection of the subassembly.

4.4.2 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

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4.4.3 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.5.3. Failure of any unit in the inspection sample to comply with 3.4 shall be cause for rejection of the assembly.

4.4.4 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.5 shall be cause for rejection of the lot.

4.5 Test methods and procedures.

4.5.1 Hydrostatic pressure. Prior to application of protective finish the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.2.

4.5.2 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.3 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.5.4 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8830994 and 8830995. Level B shall be in accordance with Drawings 8830994 and 9217624.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-731-9272.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

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6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. The responsible inspection activity (see 4.1)

d. Any special instruction requirements (see 4.3)

e. Level of packing and marking (see 5.1)

6.3 Definition.

6.3.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3.2 Exterior components. Remanufactured exterior components are: cap, firing pin housing, and chamber.

6.4 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.5 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity

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Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0D94)

Review activity:
Air Force - 70

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